at Dunsinsk Observatory, where Hamilton was stuck for the rest of his life. Hamilton ploughed through university studies by winning every conceivable honor and took his job at the Observatory even before graduation, but as a practical astronomer he proved to be a failure. Tedious observations did not appeal to him; he found theoretical subjects much more exciting.

Hamilton's social circle included major Romantic poets and philosophers. Hamilton himself harbored poetic aspirations, but William Wordsworth gently channeled his creative energies back to mathematics. Hamilton immersed himself in the reading of Kant, absorbed the Kantian notions of space and time as pure intuitions, and became intrigued by Kant's casual remark that just as geometry was the "science of space", algebra could be thought of as the "science of pure time" [**61**]. Later Hamilton insisted that Kant merely confirmed his own ideas and reading him was more "recognizing" than "discovering" [**17**, 2:96, 2:98].

Hamilton made his reputation as a mathematician by his studies in optics and mechanics, based on his notion of the characteristic function. He saw his greatest achievement in reducing the solution of $3n$ ordinary differential equations of the second order to the solution of two partial differential equations of the first order and second degree. It was not obvious that this represented any progress toward actually solving the problem, but Hamilton was convinced that even if "no practical facility is gained" from his method, the reduction of all complex calculations to the study of one characteristic function should give one "an intellectual pleasure" (quoted in [**18**, p. 89]).

Hamilton divided all algebraists into three schools of thought: the practical, the philological, and the theoretical. The practical school viewed algebra as an art and was interested in computation; the philological school viewed it as a language, a set of symbols to be manipulated according to some rules; and only the theoretical school, to which Hamilton modestly assigned himself, treated algebra as a science, "strict, pure, and independent, deduced by valid reasonings from its own intuitive principles" [**17**, 2:48].

"I am never satisfied unless I think that I can look beyond or through the signs to the things signified", wrote Hamilton. He was